

Sokhda kharif 2025: a final yield forecast from six Capella X-band passes

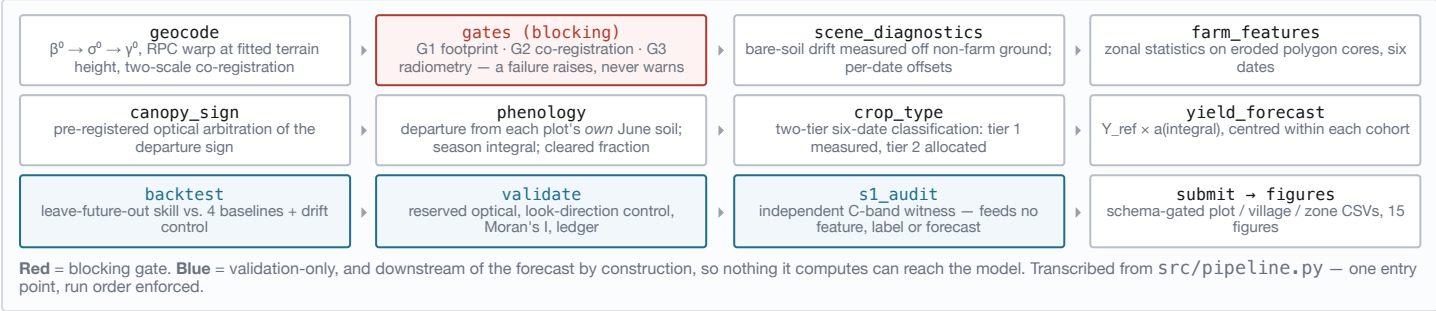
966 farm plots, one village, no ground truth — so validation is the deliverable.

Project Title: Sokhda kharif 2025 final yield forecast (SAR-only)
Primary Domain: Remote Sensing — Precision Agriculture / Crop Yield Forecasting
Data Sources Used: Capella C14 stripmap HH SLC x6, 6 Jun–12 Nov 2025 (competition) · Sentinel-2 L2A via Earth Search · Sentinel-1 RTC via Microsoft Planetary Computer (validation only) · NASA POWER daily precipitation 1995–2025 · DA&FW 3rd Advance Estimates 2025-26 · CGWB Vadodara brochure. Every external source is free and re-obtainable without registration.

Team Name: _____
Track: Round 3, ANRF AISEHack 2.0 · Sokhda, Vadodara, Gujarat

1. Proposed Strategy & Technical Novelty

The problem, stated exactly. Forecast *final yield at harvest* for each of 966 plots from six Capella X-band HH passes spanning 6 Jun – 12 Nov 2025, plus a village rollout by crop. The median plot is 0.27 ha. There is **no ground truth, no leaderboard and no scored metric** — a panel scores the submission against a published rubric. That inverts the usual priority: with no label to fit, model capacity buys nothing that can be checked, so the effort goes into controls, held-out tests and an honest uncertainty budget instead. Everything below follows from that one fact.



AI methodology — a deterministic, physics-informed chain, not a learned regressor

$$Y_{final}(\text{plot}) = Y_{ref}(\text{crop}, 2025\text{--}26) \times a(\text{season-complete canopy integral})$$

A deliberate choice, not a shortcut. With no ground truth and no scored metric on this AOI, a CNN has nothing to regress onto and an ensemble has no out-of-fold estimate to weight by — a learned model here would be fitted to its own assumptions and could not be caught doing it. Round 2 measured that failure directly: within a cohort, its health index and its yield estimate correlated at **exactly 1.000** — one ranking wearing two names. So Round 3 ships **one** modulation term and fewer modules than Round 2, not more. The radar *rank*s plots inside a crop cohort; the published state yield sets where the cohort sits.

Novel contribution — the twist

- A pre-registration ledger, shipped as a figure.** Seventeen claims were written into the code before the data that could test them was opened: **nine contradicted, one not met, seven held**, and no entry edited afterwards to agree with a later measurement. Four contradictions deleted a term, a rule or a whole module. With no labels, a record of what the model predicted before it looked is the only thing that can keep it honest.
- Each plot referenced to its own June bare soil**, never to an absolute σ^0 threshold, so plot size, soil texture and row orientation cancel in the difference — the within-image self-reference is what makes six scenes comparable at all.
- A hostile scene handled by refusing to correct it.** T5 (29 Oct) is **right-looking** at view azimuth 318.4° where the rest view from ~135°, a 01:37 IST pass three days after 63.1 mm of rain. Its radiometric residual *changes sign* with brightness (−3.5 dB dark, +2.3 dB bright): rain brightens rough soil while the reversed look extinguishes the dihedrals that make built-up bright. No constant undoes two opposite mechanisms, so **T5's level is never used** — it is replaced by the T4–T6 interpolation and only its residual survives as a weak covariate.
- An independent instrument as a witness, not a feature.** Nothing in the competition stack observes the window where the model holds cotton flat. Sixteen free Sentinel-1 RTC passes do. They are read *after* the forecast is computed, and a test asserts no model module imports the audit — a leaked column would move the village total.

Success metrics — RMSE / MAE / R² on yield cannot be computed by anyone here

No observed yield exists for these 966 plots, so no error metric against yield is available to us or to the panel. Rather than report a proxy dressed as accuracy, the work is scored on five things that *can* be measured, each of which can fail:

- Leave-future-out skill** against persistence, with the district bare-soil drift handed to every predictor so no rule can win by being biased in the drift's direction. It is the headline result, and it is negative.
- Reserved scenes.** Two optical dates were held out of the very first fetch, and an assertion fails the run if any module but the validator so much as names them.
- Spatial honesty.** A 500 m block bootstrap replaces an analytic p-value that assumes independent plots — which this project's own Moran's I says they are not.
- A null ablation.** Set $a() \equiv 1$ and re-run the whole chain, to price what the radar actually buys rather than assert it.
- Cross-instrument agreement** at cohort level against C-band, on a window X-band never saw.

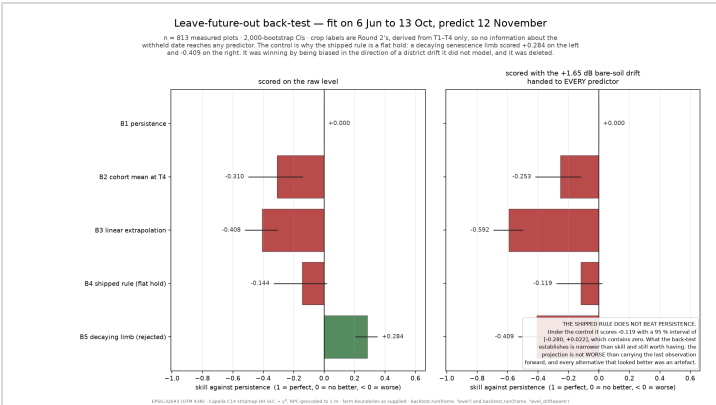
2. Ancillary Grounding & Assumptions

Assumptions the forecast rests on. Y_{ref} is the DA&FW **3rd Advance Estimates** for Gujarat kharif 2025-26 — the latest published as of 2 Sep 2026; Final Estimates for 2025-26 do not exist. No district uplift is applied, leaving maize and cotton conservative by a known sign. Cotton's canopy is held **flat, not decaying**, past the last pass. Tier-2 cohort sizes come from a district crop mix, so three of five cohorts are sized by an external prior — and the uncertainty budget carries a row for it.

Data-quality issues found and handled. The Overview promises an expanded set of villages; the shapefile holds **one** (Sokhda, ID_1 = 22), so a 500 m zone grid was added to give the rollout more than a single row. It says crop classification is carried forward; the shapefile has five fields and none is a crop, so labels were re-derived. FID is the only stable key — `id` is 1.0 on every row and one attribute has an empty name and all-null values. Ten parcels enclose under 1e-6 ha and 13 are MULTIPOLYGON; all keep a row, and area weighting stops the degenerate ones voting. The 20250619 folder holds a byte-identical duplicate of the T1 SLC, so filenames are built from the folder stem and never globbed. T6 carries a flat **+4.28 dB** sensor offset across the scene's 39 dB brightness range — corrected on invariant targets chosen from T1–T3 and scored on the two scenes that took no part in choosing them: **4.01 dB apart before correction, 0.27 after. 153 of 966 plots are not fully observed** (82 interpolated from their own dates, 71 imputed from measured neighbours), flagged per plot; the back-test scores only the 813 measured.

3. Preliminary Salient Results

Headline forecast: 893.9 t at harvest over 447.5 ha, 2.00 t/ha area-weighted, across 966 plots — 341 groundnut, 313 maize, 139 bajra, 111 rice, 62 cotton. That 447.5 ha sits inside a village boundary enclosing 1174.1 ha, so it is **38.1 % of Sokhda**; the 46-cell zone grid spreads 1.50–2.80 t/ha. Every number below is printed by the shipped run into logs/pipeline_clean.log.



The sanity check, and it is the honest one. Fit on 6 Jun–13 Oct, predict the withheld 12 Nov pass; n = 813, 2,000-bootstrap CIs; labels are Round 2's, from the first four dates only, so nothing about the withheld date reaches a predictor. **Left:** on the raw level a decaying senescence limb (B5) appears to win at +0.284. **Right:** hand the +1.65 dB district drift to every predictor and it scores –0.409 — it had been winning by being biased in the direction of a drift it did not model. It was deleted; the flat hold shipped.

QUANTIFIABLE PROGRESS	RESULT	WHAT IT ESTABLISHES
Shipped rule vs. persistence, 30-day horizon, under the drift control	–0.119 [–0.280, +0.022]	Does not beat persistence; the interval contains zero. Reported as it stands.
The same experiment at the longer horizon	+0.140 [+0.071, +0.202]	Positive at 60 days against –0.180 at 30, contradicting our pre-registration: the driver is phenology, not horizon length.
Reserved 12 Dec scene, read by no model module — cotton NDVI vs. the other four	0.690 vs 0.474–0.532, p = 1.26e–11	A SAR-only label picked the right plots on a scene it had never seen.
Canopy sign, differenced on both instruments, n = 813	rho +0.569, block bootstrap [+0.508, +0.618]	Greener is brighter here — we had pre-registered the opposite sign on four crops of five.
C-band witness — cotton over 15 Nov – 21 Dec, the window held flat; and the 6-pass season integral against the 13-pass one	+0.985 dB rho +0.915, n = 956	Cotton is the only cohort above its own June soil after 12 Nov, and <i>rising</i> , so the flat hold is not optimistic. The second reading prices the six-date premise.
Null ablation — set a() ≡ 1 and re-run the whole chain	910.1 t (1.8 %), median plot 11.8 %	The radar term redistributes within a cohort; it does not set the level.
Uncertainty budget, priced by re-running the chain under each source	external 150.9 t vs radar 9.5 t	Reference ±89.4 t, crop mix ±61.5 t, labels ±6.8 t, speckle ±2.7 t. Somebody else's numbers dominate.

Validation note. The template asks for a predicted-vs-ground-truth plot and an MAE/R² on held-out plots. **Neither exists for this AOI, and we are not going to manufacture one.** Every number above is instead a *temporally* held-out split — the 12 Nov SAR pass and the 12 Dec / 16 Jan optical scenes are withheld from the fit and read only by the validator — or a comparison against an independent instrument. Where the result is negative, as the 30-day skill is, it is printed negative.

4. Technical Challenges & Pivots

- T5 broke co-registration at 108 m** against Capella's ~5 m CE90. Diagnosed before fixing: its correlation peak is a third of the stack's and nearly flat, because at 1 m the edges a phase correlator keys on *are* shadow and layover, which swap sides under a reversed look. A two-scale search registers all six scenes to 0.06–1.48 m.
- The pre-registered canopy sign was wrong on four crops of five.** {Rice +1, Cotton –1, Maize –1, Bajra –1, Groundnut –1} was written into a constant above the code that opens the optical file; measured, it is **+1 on all five**. The sign-agnostic [departure] fallback scored –0.085 — not conservative, empty — so phenology.py was rebuilt around the signed departure, which also fixed a degeneracy that left 80.6 % of bajra on the cohort median.
- A promised per-plot harvest date was refused by its own control.** Plots the SAR called "standing" were the *least* green (one-sided p = 1.00), so the categorical label was deleted for a continuous cleared_fraction, which does validate against Sentinel-2 (rho = –0.529, n = 479).
- Reproducibility found two defects, and the second was ours.** Kaggle reproduced every SAR number and then disagreed about 39 plots: the tier-2 axis had been the November canopy clipped at zero, so 403 of 793 plots sat at exactly 0.0 and sort order alone decided the cut. We reported the fix confirmed — and our own audit found the ANOVA still residualising against the pre-fix axis. Corrected, tier-2 labels carry no optical information beyond their own axis (η² 0.0023, p = 0.43) while the tier-1 control does. Hence tier 2 is an *allocation*.
- Constraints, stated.** Tier-1 measured labels cover 26.5 % of area, **below Round 2's 31.6 %** — missed, and not met by loosening a threshold. Median peak canopy departure is 0.77 dB: real, corroborated, small. Plot-level irrigation could produce the same green-and-bright correlation we read as canopy; only the scene-level version is ruled out.

5. Final Sprint Roadmap

Immediate (next 10 hours). Assemble the Kaggle Writeup — the ≤2000-word brief, all 15 figures with cover.png as cover, the generated notebook as a public resource — and confirm both attached datasets resolve on Kaggle (round2_crops.csv for the back-test labels, and the shipped C-band table so a judge's run needs no network). One final Kaggle execution compared line-by-line against the local run log; the last comparison matched 51 of 51 signature lines. **Refinement:** no new model capacity is planned — a decision, not a shortfall, since with no labels added capacity cannot be validated. The horizon curve says the rule fails where the harvest has already happened, so a crop-calendar prior is the one improvement the evidence points at — recorded as future work, not rushed in unvalidated.

Final output. A deterministic pipeline (src/ plus a notebook *generated* from it, so the two cannot disagree), three schema-gated CSVs, 15 figures rendered by the run from its own delivered tables, a ≤2000-word brief with every number traced to the run log, 53 passing tests, and a development log that records contradicted predictions as findings.

6. Appendix

Code. https://github.com/sumitjadhav1703/SAR-Crop_Yield_Forecasting — modules, generated notebook, tests, run log, development log.

Kaggle notebook: _____

Data. Capella C14 stripmap HH SLC ×6, 6 Jun–12 Nov 2025 (competition; EPSG:32643, RPC-geocoded to 1 m) · Sentinel-2 L2A via Earth Search v1, tiles 43QBE/43QCE · Sentinel-1 IW RTC γ⁰ via Planetary Computer, 16 passes 12 Jun–21 Dec 2025, validation only · NASA POWER PRECOTCORR at 22.4254 N, 73.1567 E, 1995–2025 · CGWB Vadodara brochure.

Ground truth: none exists for this AOI; reference yields are DA&FW five-year estimates 2021–22 to 2025–26. All external sources are free and re-obtainable without registration, as the rules require.

References. Inoue, Sakaiya & Wang (2014), *Remote Sensing* 6(7):5995 — X-band σ⁰ tracks panicle biomass best, and proposes the within-image reference our departure follows. Prashnani & Justice (2026), *Remote Sensing* 18(8):1238 — cotton's phenology is extended where cereal-legume crops compress: support for the two-tier split.